INFR11287 Advanced Topics in Natural Language Processing
Mock Examination Paper 1: Mark Scheme

This mark scheme gives indicative points. Award credit for equivalent answers that demonstrate the same understanding.

1. Low-resource MT, data, and evaluation. Total: 15 marks.

   (a) Web parallel extraction: identify Yoruba/English pages using language ID and metadata; find candidate comparable pages using URLs, document structure, links, or embeddings; align at sentence/segment level using sentence embeddings or MT-based scoring; filter noisy pairs by length, language, duplication, and alignment confidence. Award up to 3 marks.

   (b) Monolingual data: backtranslation from real Yoruba into synthetic English, then train English->Yoruba; self-training or forward translation with filtering; language-model or continued pre-training on Yoruba text; LLM-assisted synthetic pairs with terminology constraints. Quality control may include round-trip checks, length ratios, COMET/quality-estimation, terminology checks, and human review. Award up to 4 marks.

   (c) Quality selection: remove noisy, toxic, duplicated, boilerplate, or language-mixed text. Diversity selection: deduplicate and balance topics/registers so all health subdomains are represented. Importance selection: choose text most useful for medical translation, e.g. Moore-Lewis selection with a medical in-domain model. Award up to 3 marks.

   (d) Advantage: transfer from a trained encoder-decoder and shared subword units can help English-side modelling and reuse parameters. Problem: vocabulary capacity may be dominated by high-resource English/German, causing poor segmentation for Yoruba morphology or characters. Award 2 marks.

   (e) Automatic metrics: BLEU, chrF, COMET, or terminology accuracy on held-out data. Human evaluation: bilingual clinical/domain review for adequacy, fluency, terminology, and harmful errors. Ethical/social harm: mistranslated health advice, unequal quality for low-resource users, asylum/medical harm, loss of trust; mention mitigation such as expert review or abstention. Award up to 3 marks.

2. RAG and open-domain QA. Total: 15 marks.

   (a) Advantage: updateable, source-grounded, citeable, domain-specific evidence. Disadvantages: retrieval can miss relevant evidence, retrieved documents can be stale/biased/noisy, generator can ignore or hallucinate over evidence, system is more complex. Award up to 4 marks.

   (b) Sparse retrieval: strong for exact terms, rare entities, lexical matches; weak for paraphrase. Dense retrieval: stronger semantic matching and paraphrase; can miss exact rare terms and needs embedding training. Hybrid and reranking combine recall with precision, especially for multilingual and factual QA. Award up to 3 marks.

   (c) Problems: translation errors in query or answer, lost cultural/local terms, retrieval from English sources may omit Arabic-local facts, answer may be fluent but unsupported. Ethical risk: inferior service for Arabic users, misinformation, cultural erasure, health/political harms. Award up to 3 marks.

   (d) Direct non-English indexing: preserves local language content and local facts, but needs multilingual retrieval and uneven corpus quality. Translate-then-index: allows one English index and model pipeline, but introduces translation errors, cost, and loss of source nuance. Award up to 3 marks.

   (e) Retrieval: recall@k, MRR, evidence relevance. Answer: exact match/F1 for factoid QA, human factuality, faithfulness, citation accuracy, LLM-as-judge with caution. Contamination/saturation: check benchmark overlap in corpus/training data; use fresh or private eval sets; inspect saturated benchmark performance separately. Award up to 2 marks.

3. MCQ explanations. Total: 20 marks.

   (a) **B.** ROUGE is based on lexical overlap with reference summaries, so it can underrate good paraphrases. BERTScore uses contextual embeddings and can reward semantic similarity such as “film” and “movie”. It does not remove references, guarantee hallucination detection, or become human evaluation.

   (b) **B.** Natural Questions contains real user questions from Google Search paired with Wikipedia evidence and annotations. This is why it is closer to open-domain QA than manually invented reading-comprehension questions. The other options confuse it with artificial datasets or yes/no-only QA.

   (c) **C.** Moore-Lewis selection compares an in-domain language model with a general-domain language model. A sentence is useful when the in-domain model explains it better than the general model, often measured by a log-probability or cross-entropy difference. It is not simply choosing rare or diverse text.

   (d) **C.** Reference-based COMET normally scores a system translation using the source sentence, the candidate translation, and a reference translation. This lets it estimate adequacy and quality beyond surface n-gram overlap. It is not computed from training loss or the translation alone in this standard setting.

   (e) **A.** Contamination occurs when evaluation items or close variants appear in training or retrieval data. It can inflate benchmark scores because the model may memorise or have already seen the answer. It is not specific to reinforcement learning and does not necessarily reduce scores.

   (f) **B.** Fusion-in-Decoder retrieves multiple passages, encodes them, and lets the decoder combine evidence while generating the answer. This differs from closed-book QA and from single-span extractive QA. The answer need not come from one copied span.

   (g) **C.** The curse of multilinguality refers to interference and capacity/data-allocation tradeoffs in multilingual models. Adding languages can help through transfer, but fixed capacity and imbalanced data can hurt some languages. It does not mean multilingual transfer is impossible.

   (h) **B.** LM Arena style evaluation asks users or judges to compare outputs pairwise, then turns those preferences into ratings such as Elo. It captures broad preference but is not the same as exact-match QA or perplexity evaluation. The result depends on prompts, users, and judge behaviour.

   (i) **C.** Backtranslation uses real target-language monolingual data, translates it back into synthetic source text, and trains on the resulting pseudo-parallel pairs. Its value comes partly from exposing the model to fluent, in-domain target-language output. The synthetic source side can be imperfect.

   (j) **C.** WEAT measures differential association between two target groups and two attribute sets in embedding space. In this example, it asks whether male names are closer to science terms relative to female names, compared with family terms. It is an embedding-bias test, not a frequency count or MT metric.